

Supporting biodiversity conservation: OpenBioMaps

Dr. Miklós Bán

University of Debrecen, Hungary

HUN-REN-UD Behavioural Ecology Research Group, Hungary

Herman Ottó Institute, Hungary

EGI webinar series, Discover Cutting-Edge Digital Solutions in Science

A tool that integrates practical data
management, decision support and research
collaboration

EGI webinar series, Discover Cutting-Edge Digital Solutions in Science

- What are the biodiversity data?
- Why these are important for the economy?
- OpenBioMaps - Framework for data collection and management:
 - The role of virtual databases in data harmonization
 - Feedback to source databases
- Making predictions?
 - Economy models
 - Automatic data processing: data usability, data collection quality, simple models
 - Feedback on data collection

Assessment

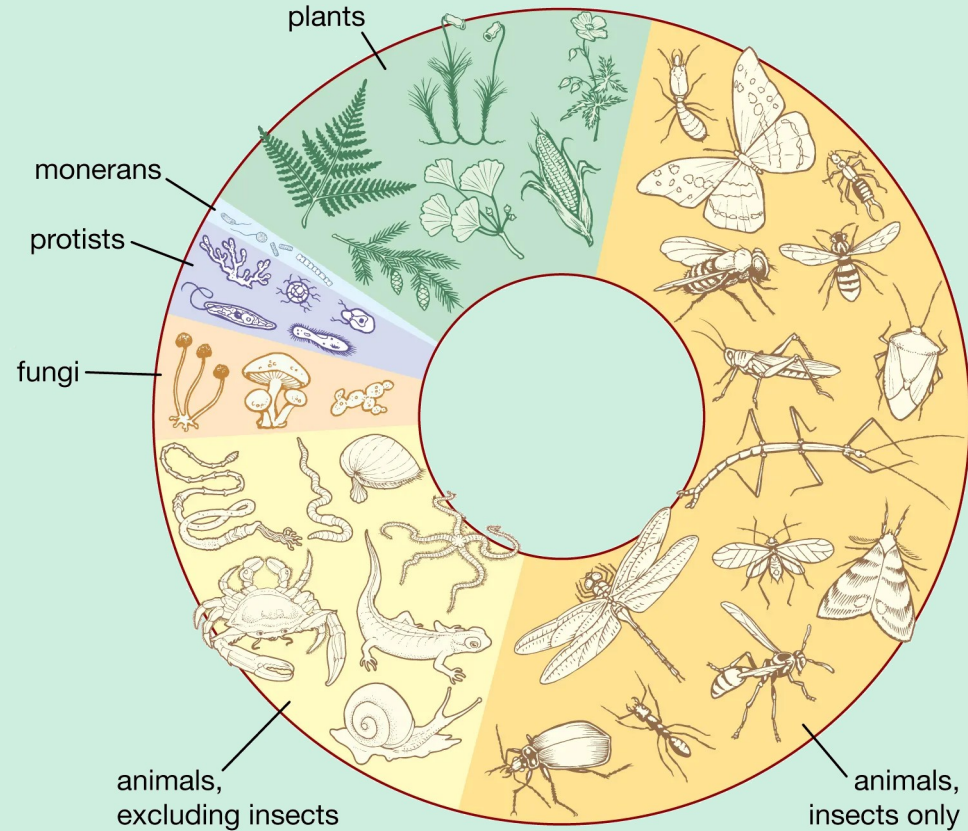
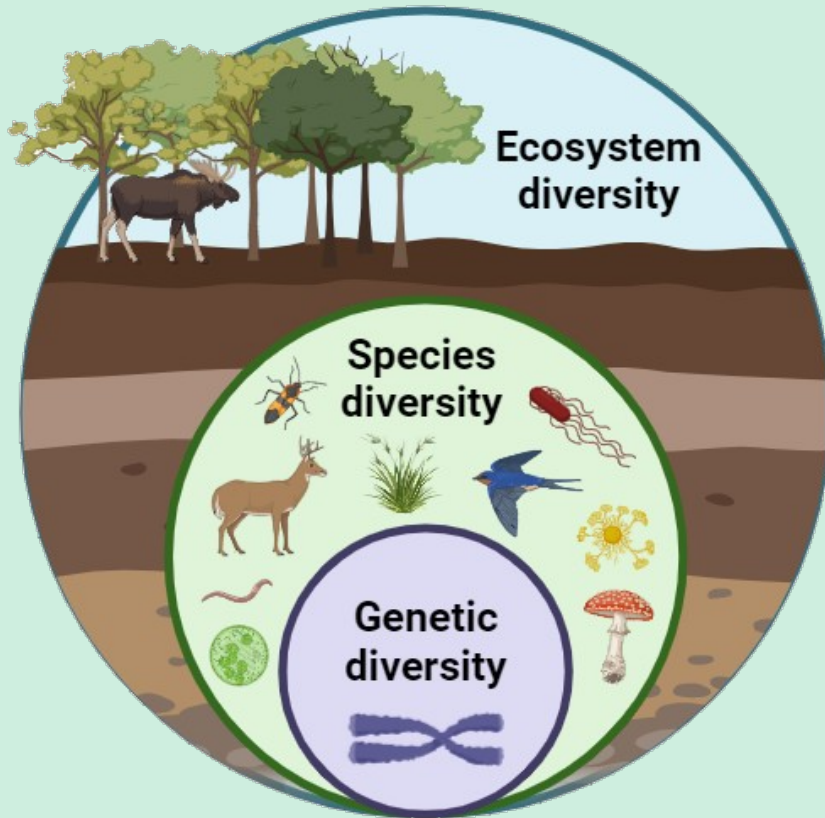
Biodiversity

Data collection

Data processing

Decision make support

Biodiversity (diversity of living nature)



Biodiversity data

- Observational data:

- What,
- When,
- Where,
- How many?

- What can we do with it?

- Species populations distribution models
- Population density estimation
- Population size estimation
- Population composition
- Track changes of a population

The economic importance of biodiversity

- **Positive effects:** soil, air, freshwater, food, climate (microclimate, biotic pump)
 - **Services:** pollination, medicine, healthy environment
- **Direct problems:** forest fires, desertification, infections, invasive species
- **Economy risks:** increasing regulations, price fluctuations, impact of epidemics, ...





OpenBioMaps

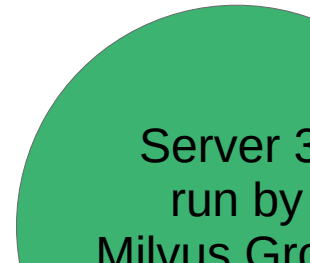
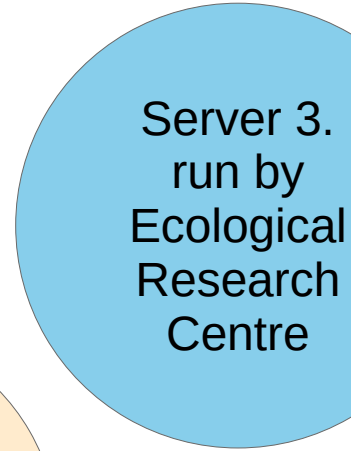
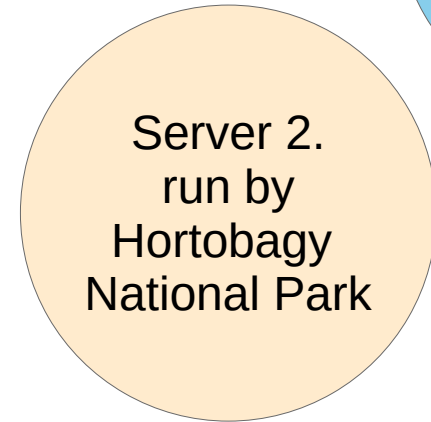
sustainable diversity

OpenBioMaps Tools

- Web maps
- User management
- Mobile apps
- R package
- Python, PHP, R extensions (e.g. for GBIF, iNaturalist,)
- Modular web interface
- Document management
- Custom app integrations
- Data visualisation
- Data processing
- Data analysis

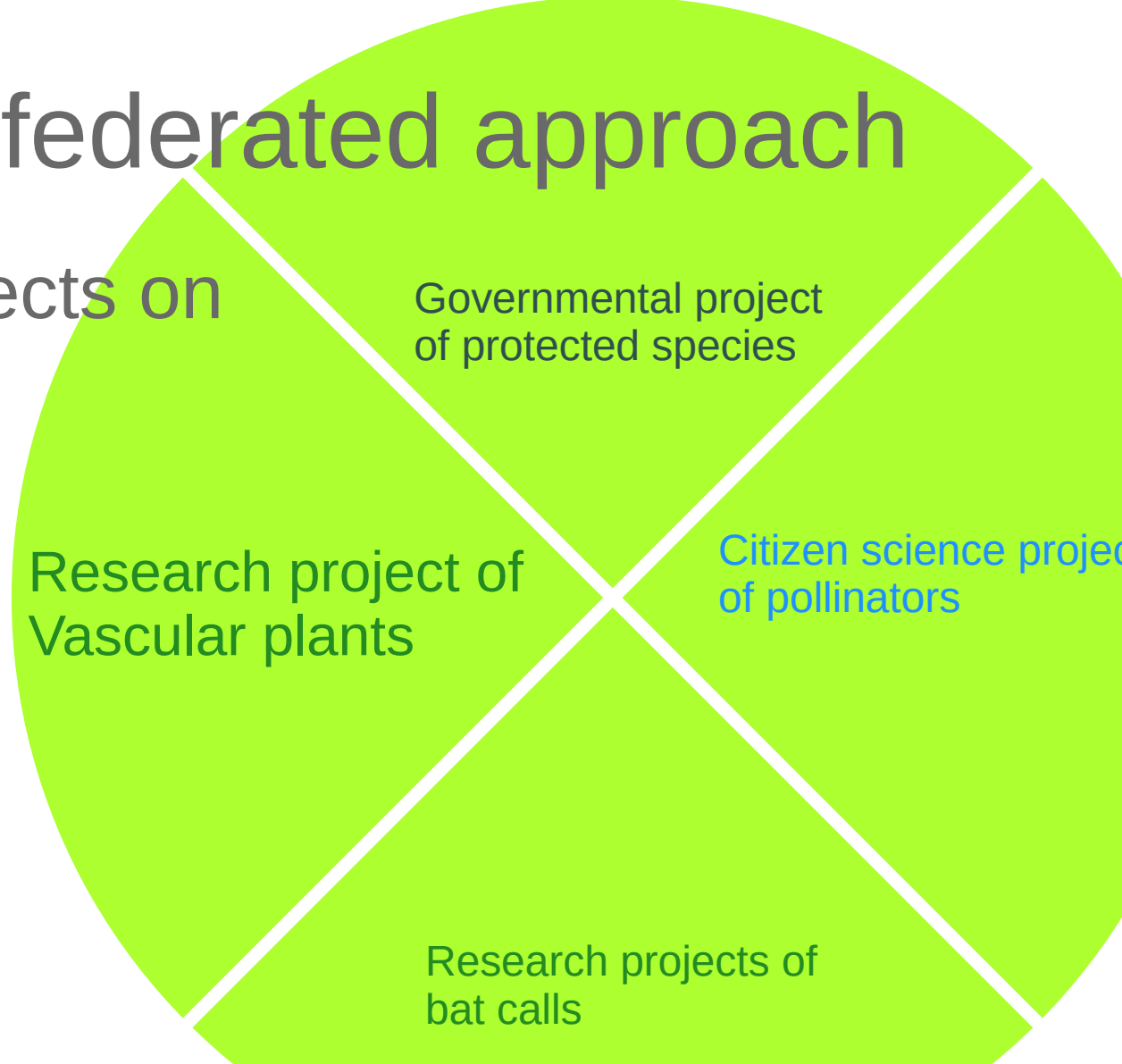
Self-hosted federated approach

- Independent servers



Self-hosted federated approach

- Independent projects on the servers



Open source approach

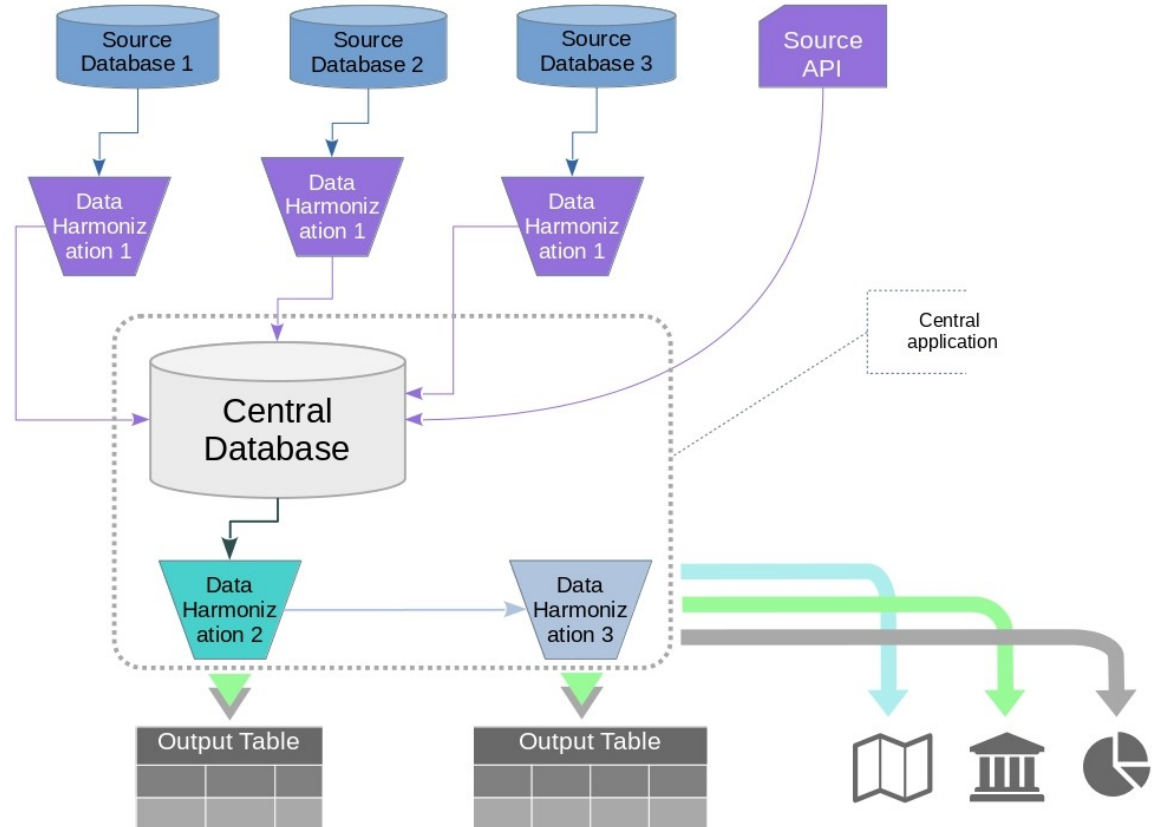
- Developer community (involve developer from different fields)
- Direct contact with users (listening them and work them together)
- CI/CD: continuous integration and continuous deployment (open source Gitlab project)
- Docker (public docker images)
- Robust open source dependencies (Postgres, Mapserver, PHP, ...)

Background services

- Sentry (app debugging)
- Weblate (community translations)
- Gitlab CI/CD (Development, integration and deployment)
- OpenBioMaps Insight (R-shiny data visualisation server)
- Jupyter (Shared analyses, collaborative work, short runs)
- “Number crunching” - R, Python computation server (Private analyses, long runs)

Data harmonisation

- Physical databases
- Virtual databases
- Usage oriented outputs



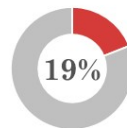
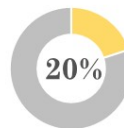
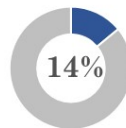
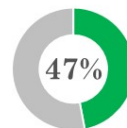
Collaboration, data processing and noise

- State nature conservation
- Citizen science
- Research institutes

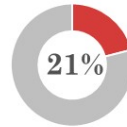
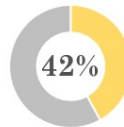
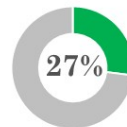
- Nature conservation
- Forest planning
- Agricultural subsidies
- National Assembly
- European green strategy



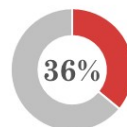
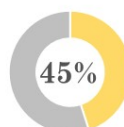
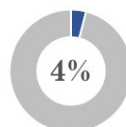
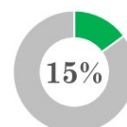
Status of bird
species
populations



Protection
status of species
other than birds



Protection
status of
habitats



Favorable Unknown Not satisfactory Bad

The status and trends of habitats and species in the
European Union in 2020



Impact indicators

What direct or indirect impact does a company or investment have on biodiversity?

- Land use, habitat change, habitat loss (e.g. deforestation and fragmentation)
- Emissions and pollution (water, soil, noise, light, air)
- Harvesting and use of natural resources
- Invasive species and biology risks



Exposure indicators

The extent to which a company is exposed to risks associated with biodiversity loss:

- Relying on natural capital
- Supply chain exposure
- Regulation risks
- Financial risks



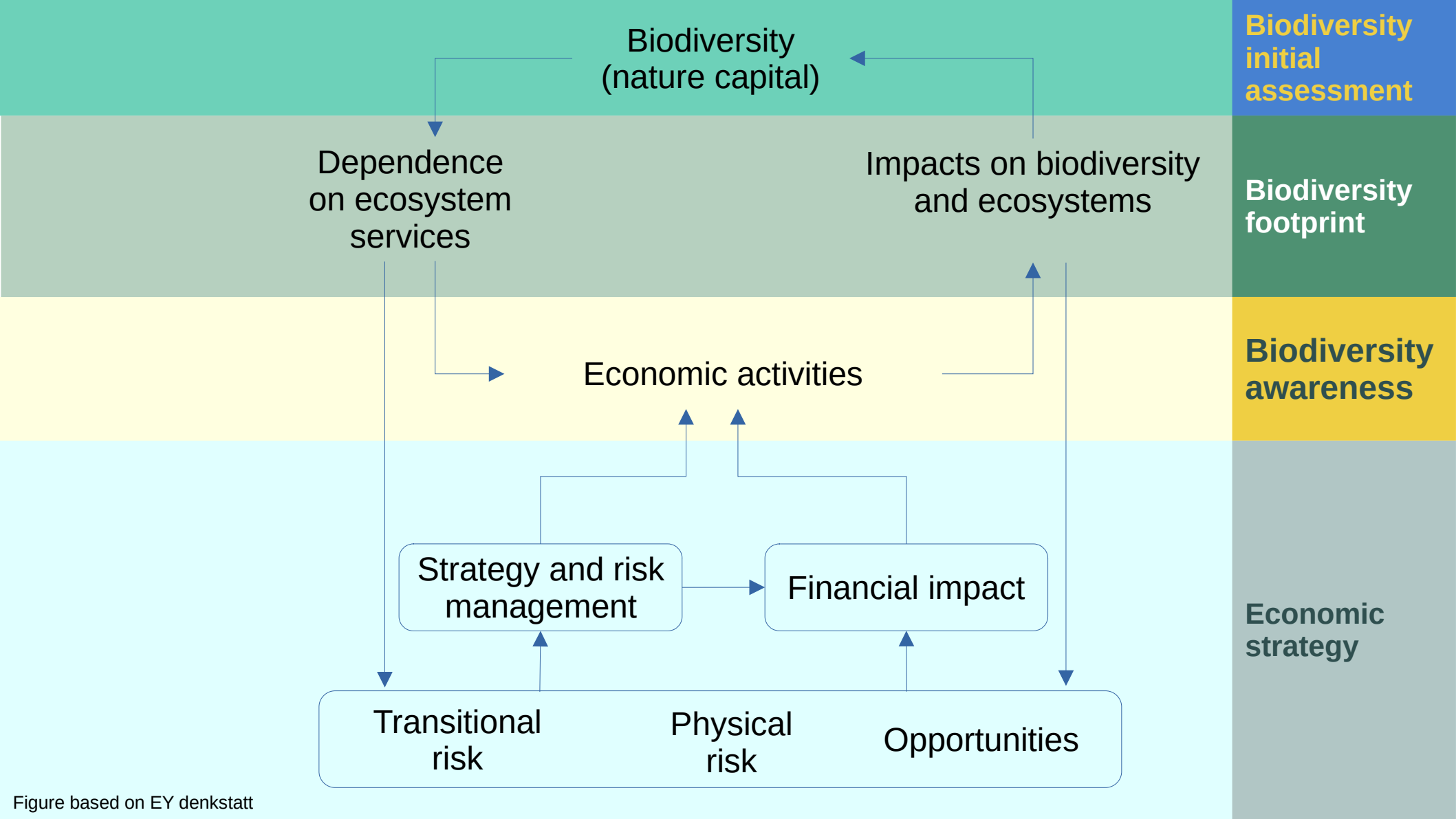


Figure based on EY denkstatt

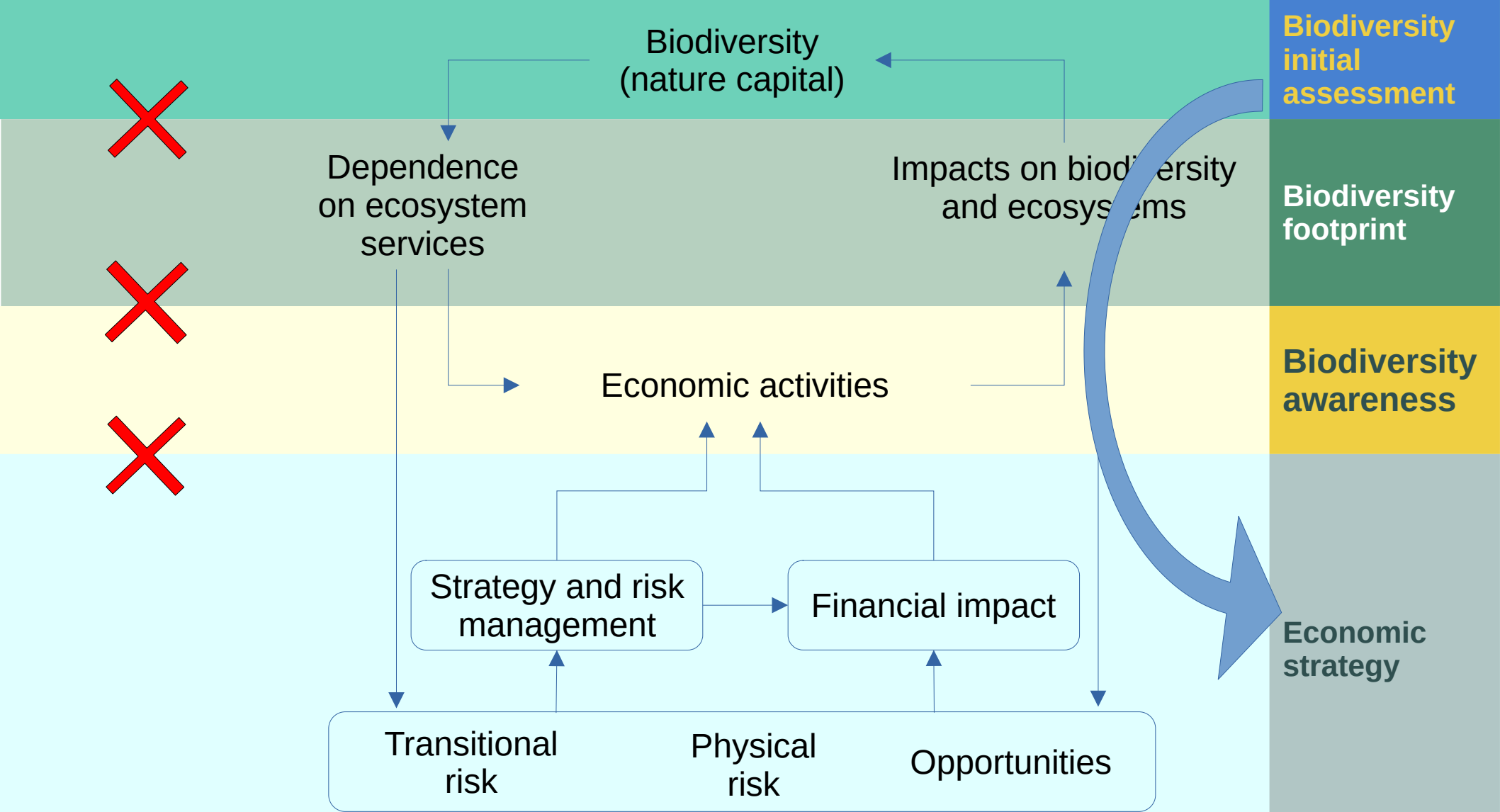


Figure based on EY denkstatt

Analysis and decision support for producers and government agencies

Predictive modelling to identify biodiversity risks

- Combining species distribution modelling and management planning scenarios to model management impacts.



Decision support for producers

- Assessment of social and political factors and restrictions.

Biodiversity stress tests for policy decisions

- Biodiversity stress scenarios (e.g., what impact would the degradation of a particular species, population, or habitat type have on public opinion).



Decision support for government institutions

- Government agencies can integrate biodiversity data into decision support systems.
- Models that convert natural capital into social value can help in planning processes on a national scale.

Analysis and decision support for business companies and financial organizations

Predictive modelling to identify biodiversity risks

- Combining species distribution modelling and economic planning scenarios to model financial impacts.



Decision support for business entities

- Assessment of social and political influences and restrictions.

Biodiversity stress tests for financial portfolios

- Biodiversity stress scenarios (e.g., what impact would the degradation of a particular species, population, or habitat type have on the assets of insurers and banks).



Decision support for financial institutions

- Financial institutions can integrate biodiversity data into decision support systems.
- Models that convert natural capital into economic value can help price sustainable investments and insurance.

Bottom-up approach

- Sharing analyses
- Sharing experiences
- Education
- Research collaborations
 - Research support for nature conservation planning
 - Research support for investment and business planning



OpenBioMaps
sustainable diversity

<https://openbiomaps.org>

Dr. Miklós Bán
banm@vocs.unideb.hu